EXCALIBUR

ENTERPRISE PORTFOLIO

STEM Racing Panama · Development Class · 2026
The Oxford School · Car: Arthur 1

Manuel Gomes
ENGINEERING · 3D CAD

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EXCALIBUR KNIGHTS

Enterprise Portfolio

STEM Racing Panama · Development Class · 2026
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Instagram @excalibur.knights

01 Team Identity

Who we are, what we called ourselves, and why it looks like this.

Team name

OUR NAME	Excalibur
WHAT IT MEANS	King Arthur's legendary blade, a symbol of courage in the face of great challenges.
WHY WE CHOSE IT	Like Excalibur, we want what we create to be recognisable as ours — built with purpose and made to leave its own mark.

Team goal & ethos

1	Learn as much as we can from the STEM Racing competition.
2	Gain hands-on experience in STEM, building skills that will help us in the future.
3	Gain experience for future competitions.

Logo development

VERSION 1 VERSION 2 FINAL

Fig. 1.1 Logo versions 1 – final

VERSION 1	Hand-drawn in Procreate, all black on white. Dropped because the checkered flag did not convey a sense of speed.
VERSION 2	Ivory background, navy wordmark, royal blue on the blade and base of the "E".
FINAL	The flag curves into the "E" for speed. Sword = strength · crown = excellence · flag = STEM Racing · "E" = our name.
SOFTWARE AND WHY	Procreate and Canva (iPad) — easy to use and already familiar to us.

Evaluation of team identity

IS IT WORKING?	Yes, the same colours, logo and story appear on the car, the posts and the sponsor material. Our "Who are we" post was the most liked, which tells us people connect with the team behind the brand.

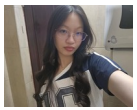
Colour palette

	Navy Blue PRIMARY · #0A2457	Trust and focus — the "E" and our text
	Royal Blue SECONDARY · #1E4ED8	Energy and speed — blade and base of the "E"
	Golden Yellow ACCENT · #F4B400	Excellence and ambition — the crown
	Ivory White LIGHT · #FAF8F2	Space and clarity — backgrounds
	Cool Grey Blue NEUTRAL · #D6E0EF	Support colour for dividers and panels

The team



Manuel Gomes
Engineering, 3D car model (CAD)



Heidy Qiu
Enterprise, marketing, sponsors & painting



Victoria Duran
Enterprise, marketing,, sponsors & painting



Roberto Ho
Engineering, simulation & assembly



Nabil Manjara
Engineering, simulation & assembly



Giancarlos Pan
Engineering, simulation & assembly

Applying the identity

CAR	Navy base, electric-blue gradient, gold accents and ivory — the sword runs down the nose and the crown sits on the rear wing (see section 05).
SPONSOR MATERIAL	Our sponsorship proposal and tri-fold leaflet use the same navy, gold and ivory, with the final logo on the cover.
SOCIAL MEDIA	Every Instagram post uses the palette and the logo, so the account reads as one brand.

02 Project Management & Teamwork

How six people organised themselves and kept to a deadline.

How we organised

TOOLS AND WHY	Google Drive — to store and share project files in one place, so everyone could access the latest version Google Calendar — to track project deadlines
HOW ROLES WERE ASSIGNED	Based on the strengths and skills each member brought: four of us focused on engineering (design, simulation, assembly) and two on marketing, sponsors and art.
HOW OFTEN WE MET, AND WHERE	Every Friday in a school classroom, with extra weekday sessions around key deadlines.

Timeline

KEY DEADLINES WE SET	20 September — final car design · 30 September — car painted and finished · 10 October — portfolio complete
WHICH ONES WE HIT	The car design and painting deadlines.
WHICH SLIPPED, AND WHAT WE DID	The portfolio deadline slipped. We had treated it as less urgent than the car, so we worked on it together as a team to finish it quickly.



Fig. 2.1 Our key milestones

A decision we made together

THE DECISION	Choosing the car's final livery.
THE OPTIONS	Three livery designs with different colour schemes and decal layouts. We liked all three.
HOW WE CHOSE	We compared each option's strengths and weaknesses and combined the best elements of the three into one final design.
WAS IT RIGHT?	Yes, the final livery gives the car a clear, consistent look that matches our identity, and everyone's ideas are in it.

Responsibility matrix

R = responsible · C = consulted · I = informed

Task	Manuel	Heidy	Victoria	Roberto	Nabil	Giancarlos
3D car model	**R**	I	I	C	C	C
Car simulation	C	I	I	**R**	**R**	**R**
Car assembly	C	I	I	**R**	**R**	**R**
Marketing & sponsors	I	**R**	**R**	I	C	I
Car painting	C	**R**	**R**	I	I	C

Communication

CHANNELS AND WHAT FOR	Gmail — official announcements and meeting arrangements Instagram — team discussions and decision-making Discord — resource sharing
ABSENT TEAM MEMBERS	After each meeting a report recapped the key decisions, assigned tasks and upcoming deadlines.
CHECKING AGAINST THE REGULATIONS	The car was checked at every design review to make sure it met all requirements, and any changes were made before testing.

What did not work

WHAT WENT WRONG	The first livery design looked too busy.
WHY	The colours and graphics competed for attention.
WHAT WE CHANGED	We simplified the design.
THE RESULT	It gave the car a cleaner look.

Working with others

TEACHERS OR EXPERTS CONSULTED	Our physics teacher, for guidance on the scientific principles behind the car's design.

Evaluation of management

DID OUR ORGANISATION WORK?	Mostly. Regular deadline reminders kept tasks organised and we hit both car deadlines. Our weak point was prioritisation: we did not treat the portfolio as urgent, and it was the only deadline we missed.

03 Marketing & Outreach

How we told people who we are.

Social media

Instagram	15	161
@excalibur.knights	posts	followers

WHY THIS PLATFORM	Instagram is where our community is most active and where we reach the most people.
WHAT WE POST AND HOW OFTEN	Team-name story, logo meaning, colour palette, sponsors and who we are — every two days or after each team meeting.
HOW WE PLANNED TO GROW IT	Posting content our community engages with, so each post reaches more people.

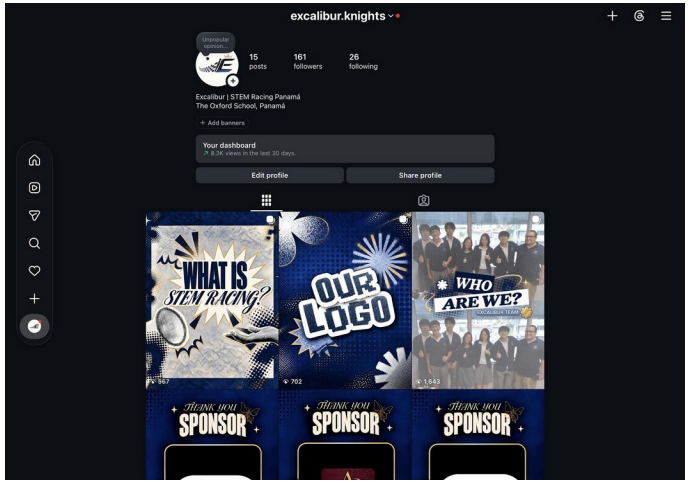


Fig. 3.1 Our Instagram profile

Website

URL	https://excalibur-knights.netlify.app/
SECTIONS	The car, engineering, our journey, the knights, and contact
WHAT WE USE IT FOR	It showcases the car and its engineering, shares information about Excalibur and provides contact details for anyone interested in the team.

What we posted



Team name
The story behind Excalibur and why we chose it.



Logo
The four elements of the logo and what they mean.



Colour palette
Our five colours with their hex codes.

Fig. 3.2 Three of our posts and what each was for

POST TYPE THAT WORKED BEST	Our "Who are we" post had the most likes — people connect with the faces behind the team.
POST TYPE THAT DID NOT	Sponsor posts had the lowest reach. Next time we will feature sponsors alongside team content.
ENGAGEMENT NUMBERS	7.1k views in the last 30 days, an average of 15 likes per post.

04 Sponsorship & Finance

Who we asked, what we offered, what it cost, and what came back.

Our strategy

TYPE OF COMPANY TARGETED	Technology companies and family-run local businesses.
WHY THAT TYPE	They are connected to our own community, so a warm introduction let us pitch in person, explain the project properly and get a faster answer than with cold emails.
HOW WE FOUND THEM	Through our own network: businesses connected to our families and school community.
WHAT WE LEARNED	A personal connection opens the door, but every sponsor still wanted to see a proper proposal before saying yes.

The approach

WHAT OUR PROPOSAL LEADS WITH	Who we are, what STEM Racing is and why the project matters to us — before any request for money.
WHAT WE OFFERED	Brand visibility on our car, uniforms and digital media; association with an international competition backed by Formula 1; partnership in the first edition of STEM Racing Panamá; and support for young talent and innovation in Panama.



Fig. 4.1 Our sponsorship leaflet (outside of the tri-fold)

Our sponsors

All nine sponsors made a financial contribution.

D&A — Durán y Asociados Financial contribution	**SISAP — Sistemas Aplicativos** Financial contribution	**Servicios y Sistemas (S2)** Financial contribution
Solusoft Financial contribution	**Ser Masivo** Financial contribution	**FerreMateriales** Financial contribution
Mini Super Mar Dorado Financial contribution	**Smile Factory** Financial contribution	**ALTAF — Perfumes y más** Financial contribution
C&P Biker Financial contribution	**Academia de Artes Marciales Leon** Financial contribution	**RICA NGEL S.A.** Financial contribution

Evaluation of sponsorship

WHAT WORKED	Starting with our own network got us nine sponsors in our first season, and every one of them received a formal proposal and leaflet.
WHAT WE WOULD CHANGE	All nine came through personal connections, so we have not yet tested our pitch with companies that don't know us. Next season we plan to approach local engineering and technology businesses directly, using this year's results as proof of what we can offer.

05 Brand Application & Evaluation

Where the brand ended up in the real world, and what we make of it all.

The car



Fig. 5.1 Final livery with sponsor and STEM Racing decals (top and side view)

HOW THE PALETTE WAS APPLIED	Navy as the base, an electric-blue gradient for speed, gold accents linked to the crown in our logo, and ivory for contrast and legibility.
SPONSOR LOGOS — WHERE AND WHY	The same layout on both sides of the car. Dark logos were recoloured white (and ALTAF gold) so they stay legible on navy; colourful logos sit on the ivory band in their original colours. The top view mirrors left and right.
STEM RACING DECALS (D4.5)	Original white decal on the middle sidepod of each side, covering the first third and flush with its front edge.

Producing the graphics

SOFTWARE AND EQUIPMENT	Canva and Procreate (iPad) for design; logos exported as transparent PNGs for printing.
A PROBLEM WE HIT	Several sponsor logos (dark blue or black) disappeared against our navy paint.
HOW WE FIXED IT	We made light versions — white, or gold for ALTAF — and kept the original colours only on the ivory areas.

Brand guidelines

LOGO	Full-colour logo on ivory or white backgrounds; never stretched or recoloured.
COLOURS	Navy, royal blue, gold and ivory only — the same five hex codes on the car, posts and print.
SPONSOR LOGOS	White or gold on navy, original colours on ivory, always on transparent backgrounds.

Overall evaluation

Our final reflections

WHAT WE ARE PROUDEST OF	How well we worked as a team and how good our communication became.
OUR BIGGEST MISTAKE	Not treating some objectives as important or urgent — especially the portfolio.
WHAT WE WOULD DO DIFFERENTLY	Work more efficiently: set clear priorities from the start and treat every deliverable as part of the competition.
WITH THREE MORE MONTHS...	We would finish the portfolio earlier and use the extra time to test the car more and give our sponsors more visibility.